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The Costs of Information Processing in an Information Ratchet System

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The Landauer's Principle states that the erasure of a bit of information must dissipates $k_B \ln 2$ amount of entropy. However, Landauer's Principle failed to account for the structural costs associated with the implementation of the information processing device. These structural costs can result in inefficient processing of information. By studying the information ratchet system, which mediates interactions between heat, work, and information reservoirs, the relationship between information and energy flows can be quantified. The existence of an unsurpassable bound of k_B/e that limits efficient is found beyond the bound of Landauer's Principle. Such structural costs are found to come from dissipations of unaccounted correlations formed between the device and the information reservoir, known as the transient and modularity dissipations.

I. INTRODUCTION

Modern physics of information tells us that there is a cost associated with information, that other than the creation and storage of information, even the act of erasure requires energy to be expended. This is the famed Landauer's Principle [1], which quantified the costs associated with the burden of information. Together with the development of information theory by Claude Shannon, information is seen as a physical quantity, and plays a part in thermodynamic systems.

A modified Second Law was proposed that accounts for the informational entropy of a thermodynamic system [2]:

$$\langle \Delta S \rangle + k_B \ln 2 \ \Delta H \geq 0 \tag{1}$$

Whereby ΔS and ΔH is the change in thermodynamic entropy and the change in Shannon information entropy respectively. The Landauer's Principle is apparent in this modified Second Law—an erasure of 1 bit of information corresponding to $\Delta H = -1$, will guarantee an increase in entropy of at least $k_B \ln 2$:

$$\langle \Delta S \rangle \geq k_B \ln 2 \tag{2}$$

However, the Landauer's Principle does not capture any additional costs from the device doing the erasure. One would expect different device's implementations to have different efficiencies of erasing. Eraser devices might also become correlated with the system, hoarding some amount of information or energy. To quantify the costs of eraser devices, we will study an effective and widely used model—the information ratchet [3, 4], which uses a tape of string as its information reservoir. The information ratchet is useful as it is a device that directly mediates interactions between the heat, work and information reservoirs. The creation or erasure of information in the information reservoir is coupled to the energy flows between the device and the heat and work reservoirs. The study of ratchet efficiencies have also been the focus of recent studies [4–6].

As the main result of this letter, numerical simulation of the most efficient information ratchet obtained a stricter bound to the modified Second Law that cannot be surpassed, implying additional costs associated to the ratchet's implementation. Information-theoretic analysis of the information ratchet system revealed the sources of these additional costs as erasures of correlations between the ratchet and the information reservoir, known as the transient dissipation, and between the interacting and stationary subsystems, known as the modularity dissipation.

These results suggest that more thoughts should be taken in designing information processing devices to minimise energy costs. It is also important to note that the existence of these costs are not an engineering problem. Rather, they are a fundamental limit similar to the Landauer's Principle. The study of the information ratchet also hints at multiple possible future applications due to their similarity to a Turing machine, as well as DNA polymerase or other protein motors [7, 8].

II. THE INFORMATION RATCHET

The information ratchet is an abstract physical model of a ratchet moving unidirectionally along a tape of binary strings as shown in Figure 1. The system consists of 4 parts: An infinite tape of symboled strings serving as the information reservoir, a mass depicted as being lifted by a pulley as the work reservoir, a heat reservoir at a temperature T, and a finite-state ratchet that governs and mediates interactions between all three reservoirs.

Travelling along the tape, the ratchet exchanges energies with the heat and work reservoirs, while writing or erasing information on the tape by changing the individual symbols the ratchet is on. The ratchet will extract heat from the heat reservoir, converting it to work, seemingly violating the Kelvin-Planck statement of the Second Law of Thermodynamics. This Maxwell's Demon engine operation is only possible with the inclusion of the information reservoir, whereby the entropic costs required by the Second Law are offset by the change in entropy of the information reservoir. The finite-state ratchet has a finite number of internal states $\mathcal{X}$, similar to the internal states of a Turing machine, while the symboled tape have symbols $\mathcal{Y} \in \{0, 1\}$. At each interval,

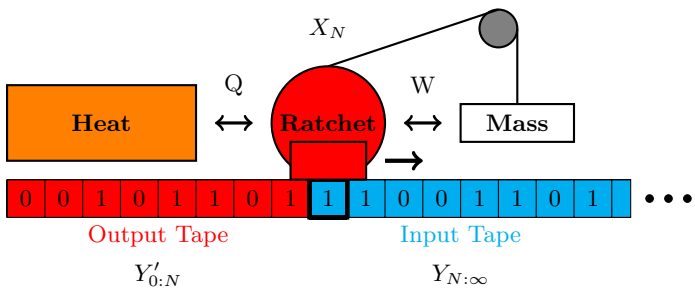

FIG. 1. Information ratchet on the N_{th} symbol with ratchet state X_N, exchanging heat Q and work W, while moving along the input tape $Y_{N:\infty}$ and writing new symbols to the output tape $Y'_{0:N}$.

the ratchet is on a single symbol on the tape, resulting in a joint ratchet-tape states of $\mathcal{X} \otimes \mathcal{Y}$. For example, a ratchet with two internal states A and B will be referred to as a 2-states ratchet, and will have:

$$\begin{aligned} \mathcal{X} &\in \{A, B\} \\ \mathcal{Y} &\in \{0, 1\} \\ \mathcal{X} \otimes \mathcal{Y} &\in \{A \otimes 0,\ A \otimes 1,\ B \otimes 0,\ B \otimes 1\} \end{aligned}$$

The ratchet will start at the very first symbol of the tape Y_0 with an initial ratchet state X_0, having a joint state of $X_0 \otimes Y_0$. The ratchet will then interacts with the heat reservoir for an interval τ, and transiting to the joint state $X_1 \otimes Y'_0$ (interaction transition). This means that the ratchet state transits to X_1 and changed Y_0 to Y'_0 at time $t = \tau$. The ratchet will then interacts with the work reservoir and move on to the next symbol Y_1, forming the new joint state of $X_1 \otimes Y_1$ (switching transition), and the same process continues.

Going through this process from Y_0 until Y_N at $t = N\tau$ will output a new sequence of symbols $Y'_{0:N}$ (shown in red in Figure 1), whereas the yet to be processed sequence is $Y_{N:\infty}$ (shown in cyan). The interaction transition from $X_N \otimes Y_N \to X_{N+1} \otimes Y'_N$ is governed by a Markov transition matrix M:

$$M\vec{\rho}(N) = \vec{\rho'}(N) \tag{3}$$

Where $\vec{\rho}(N)$ and $\vec{\rho'}(N)$ are the probability state vectors of the joint states before and after transition respectively. M has elements of $M_{x_N \otimes y_N \to x_{N+1} \otimes y'_N} = \Pr(X_{N+1} = x_{N+1}, Y'_N = y'_N \mid X_N = x_N, Y_N = y_N)$. This means that each element of the matrix denotes the probability of transition from one state to another, and must be positive. Note that the transition matrix M is a left-handed matrix, such that its column elements sum up to unity (completeness) instead of the row elements. M must also be a square matrix, with its number of columns being the length of $\mathcal{X}$ multiplied by the length of $\mathcal{Y}$.

These ratchet's energetics and transition probabilities of the system are fully captured in the Markov transition matrix M by Boltzmann statistics:

$$\begin{aligned} \frac{M_{x_{N+1} \otimes y'_N \to x_N \otimes y_N}}{M_{x_N \otimes y_N \to x_{N+1} \otimes y'_N}} &= e^{(E_{x_{N+1} \otimes y'_N} - E_{x_N \otimes y_N})/k_B T} \\ \implies \Delta E_{x_N \otimes y_N \to x_{N+1} \otimes y'_N} &= k_B T \ln \frac{M_{x_{N+1} \otimes y'_N \to x_N \otimes y_N}}{M_{x_N \otimes y_N \to x_{N+1} \otimes y'_N}} \end{aligned} \tag{4}$$

All Markov transition matrices M must satisfy equation 4 as the ratchet is thermalised with the heat reservoir, with their transition probabilities dependent on the energy difference of the transitions. Each unique ratchets are then fully described by their unique transition matrix M.

The interaction transitions are governed by M and equation 4, and takes place in an interval of τ. This is when the ratchet interacts with the heat reservoir, extracting the heat Q necessary to make the transition to different energy states. During the switching transition, the ratchet only interacts with the work reservoir, extracting work W necessary for the switching of the bits. Therefore,

$$Q_N = E_{x_{N+1} \otimes y'_N} - E_{x_N \otimes y_N} \tag{5}$$

$$-W_N = E_{x_{N+1} \otimes y_{N+1}} - E_{x_{N+1} \otimes y'_N} \tag{6}$$

It is important to note that the system gains energy when it extract heat from the heat reservoir to make a transition, but the consumed work to make the switching are lost and dissipated back to the heat reservoir, hence the negative sign. Since this is a closed system, all the work that are needed to drive the ratchet comes directly from the heat reservoir with no input from external sources, making the ratchet autonomous. Thus, intuitively it is clear that the average or asymptotic rate of heat extraction is equal to the work production:

$$\langle W \rangle = \langle Q \rangle \tag{7}$$

In the context of the information ratchet, the modified Second Law is expressed as:

$$\langle W \rangle \leq k_B T \ln 2\ \Delta H \tag{8}$$

To account for correlations, or orderings in the information tape, a new entropic measure known as the metric entropy rate or the Kolmogorov-Sinai entropy rate h_μ [9–11] is required.

$$\begin{aligned} h_\mu &= \lim_{l \to \infty} \frac{H[Y_{0:l}]}{l} \\ &= \lim_{l \to \infty} H[Y_l \mid Y_{0:l}] \end{aligned} \tag{9}$$

Thus, the new information processing relation is given in equation 10 with h'_μ denoting the entropy rate of the output tape. This is known as the Information Processing Second Law (IPSL). It has been demonstrated that the IPSL is indeed a better bound compared to equation 8 [4].

$$\begin{aligned} \langle W \rangle &\leq k_B T \ln 2\ (h'_\mu - h_\mu) \\ &= k_B T \ln 2\ \Delta h_\mu \end{aligned} \tag{10}$$

III. RESULTS

A. Achievability of Bounds

Numerical simulation of a ratchet with a single state and an input tape generated by a bias coin flip without temporal correlations, showed that our ratchet cannot achieve a work production rate of more than k_BT/e (TABLE I). It can be proved that this ratchet and tape choice is the most efficient information ratchet system as shown in appendix A.

Bias	0.5	0.6	0.7	0.8	0.9	1.0
$\langle W\rangle_{\max}/k_BT$	0	0.0183	0.0679	0.144	0.244	0.368
Efficiency at $\langle W\rangle_{\max}$	0	0.910	0.832	0.763	0.683	0.560

TABLE I. Maximum work production rates and corresponding efficiencies for the 1-state memoryless ratchet with input processes of different bias values. Values are 0 for bias 0.5 as there are no engine regimes for bias 0.5.

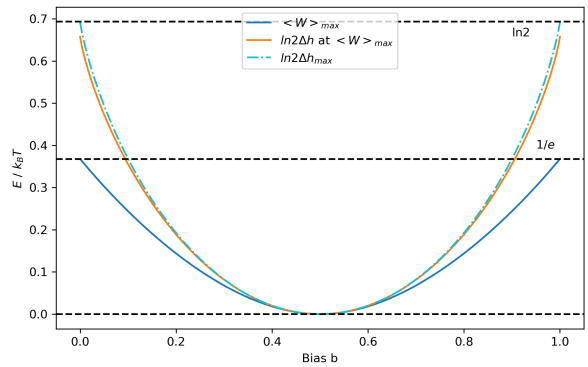

FIG. 2. Plots of maximum work production rate $\langle W\rangle_{\max}$, its corresponding $\ln 2\ \Delta h_\mu$, and the maximum possible $\ln 2\ \Delta h_{\max}$ against input process bias b. Work production rate cannot exceed k_BT/e, and the Landauer limit of $\ln 2$ is not reached.

We have defined efficiency as the achievability of the IPSL bound:

$$\text{Efficiency} = \frac{\langle W\rangle}{k_BT\ln 2\ \Delta h_\mu} \tag{11}$$

The maximum work production rates $\langle W\rangle_{\max}$, and its corresponding $\ln 2\ \Delta h_\mu$ for each information tape generated by coin flips of bias b are plotted in Figure 2. $\langle W\rangle_{\max}$ is highest at biases $b = 0$ and $b = 1$ as the information entropy of the information tape is 0 at these biases. This means that the ratchet should be able to to fully create $\Delta h_\mu = +1$ at these biases, and a $\langle W\rangle_{\max} = k_BT\ln 2$ should be obtained. However, not only is $\langle W\rangle_{\max} = k_BT/e < k_BT\ln 2$, the maximum Δh_μ is also lesser than 1 (shown in orange). These suggest there exists some fundamental costs associated to the working of the ratchet that prevents the IPSL bound to be achieved.

B. Transient Dissipation

The IPSL comes with a caveat of the complete disregard for the information or entropy that the ratchet might carries—Δh_μ is the only part of the equation that describes information, and it only consists of the change in entropy rate of the tape. However, a ratchet can and will store and process information. A more general bound that includes the change in entropy of the ratchet is:

$$\frac{\langle Q_{0\to N}\rangle}{k_BT\ln 2} \geq H[X_0, Y_{0:\infty}] - H[X_N, Y'_{0:N}Y_{N:\infty}] \tag{12}$$

Equation 12 gives the heat dissipation of processing the input process from time step 0 to N, and it is bounded by the change in entropy of the entire joint system of ratchet and tape at time step 0 and N. Unlike the IPSL, $\langle W\rangle$ is expressed as $-\langle Q\rangle$ here, as we are interested in the dissipation rather than the work production.

The IPSL can be expressed into the same form as equation 12:

$$\lim_{N\to\infty}\frac{1}{N}\frac{\langle Q_{0\to N}\rangle}{k_BT\ln 2} \geq -(h'_\mu - h_\mu) \tag{13}$$

As equation 12 accounts for the entire ratchet and tape system, and equation 13 only accounts for the tape, their difference is thus the dissipation that comes from the ratchet. We would expect that this dissipation is dependent on the ratchet's design, an efficient ratchet would have a smaller dissipation as compared to an inefficient ratchet. This dissipation is referred to as the transient dissipation [12]:

$$\frac{\langle Q^{\text{tran}}_{0\to N}\rangle}{k_BT\ln 2} \geq \lim_{N\to\infty}\left[\frac{\langle Q_{0\to N}\rangle}{k_BT\ln 2} - (-N(h'_\mu - h_\mu))\right] \tag{14}$$

Which can be expressed as equation 15 as shown in Appendix B:

$$\frac{\langle Q^{\text{tran}}\rangle_{\min}}{k_BT\ln 2} = I[X_N; Y'_{0:N}] - E' \tag{15}$$

Whereby $I[X_N; Y'_{0:N}]$ is the mutual information between the ratchet and the past outputs, and E' is the output's excess entropy [13]. Thus, an efficient ratchet must keep as little memory about the past outputs as possible to keep the mutual information small. We can interpret this as the ratchet keeping useless information about the past outputs that are unnecessary, creating and keeping useless entropy.

To understand what the transient dissipation describes, we can visualise it by drawing an information diagram shown in Figure 3.

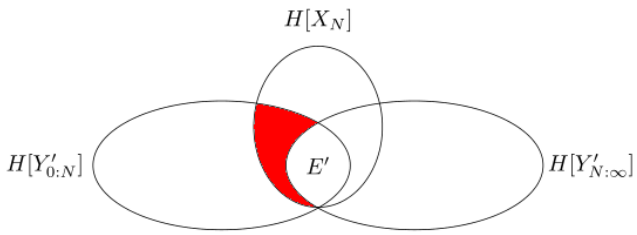

FIG. 3. Information diagram of the ratchet and output process. E' is the mutual information between the output's past $Y'_{0:N}$ and output's future $Y'_{N:\infty}$. The red area shows the transient dissipation, it is the mutual information between the ratchet X_N and the output's past $Y'_{0:N}$ subtracted by E'.

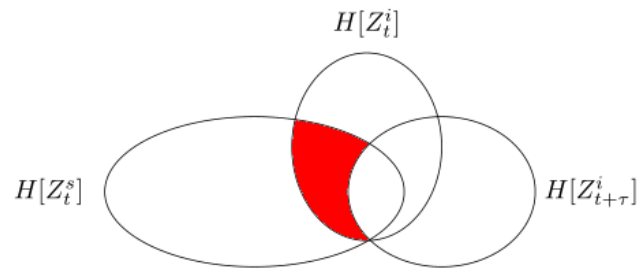

FIG. 4. Information diagram of the interacting or local subsystem and the stationary subsystem. The red area shows the modularity dissipation. This is the mutual information lost between the local subsystem and the stationary subsystem when the local subsystem transits from $Z^i_t \to Z^i_{t+\tau}$.

The information diagram simply depicts an output process with temporal correlations that are 'bridged' by the ratchet, making the ratchet a generator of correlations, or simply a pattern generator. The transient dissipation in equation 15 is thus the heat dissipation required to generate this pattern from a patternless or memoryless input tape. It make sense that the excess entropy reduces the transient dissipation, as a large excess entropy means that we can predict more future outputs from the known past outputs, and in some sense, saving the ratchet of some work.

C. Modularity Dissipation

It has been argued in [14] that information processing on individual parts of the system instead of the whole, contributes to entropy production known as the modularity dissipation. This is true for our information ratchet system, as during each interaction interval, we are only changing a single bit of the whole information tape.

Recall that during the ratchet's interaction transition, we have $X_N \otimes Y_N \to X_{N+1} \otimes Y'_N$, while the rest of the input and output tapes, $Y_{N+1:\infty}$ and $Y'_{0:N}$, remains unchanged or stationary. Therefore, we have:

$$\begin{aligned} Z^i_t &= (X_N, Y_N) \\ Z^i_{t+\tau} &= (X_{N+1}, Y'_N) \\ Z^s_t &= (Y_{N+1:\infty}, Y'_{0:N}) \end{aligned}$$

Whereby Z^i_t and $Z^i_{t+\tau}$ is the interacting subsystem before and after interaction at time t, and Z^s_t is the stationary subsystem at time t. As a consequence of the Data Processing Inequality [13], operations on the local interaction subsystem will decrease correlations between the interacted subsystem and the rest of the global stationary subsystem. Again, we can better visualise this by drawing an information diagram as shown in Figure 4.

Thus, the modularity dissipation is the change in mutual information between the interacting subsystem and the stationary subsystem, before and after the interaction of interval τ. The information-theoretic derivation is presented in Appendix C.

$$\begin{aligned} \frac{\langle \Sigma^{\text{mod}}_N \rangle_{\min}}{k_B \ln 2} &= I[X_N, Y_N; Y_{N+1:\infty}, Y'_{0:N}] \\ &\quad - I[X_{N+1}, Y'_N; Y_{N+1:\infty}, Y'_{0:N}] \end{aligned} \tag{16}$$

Therefore, after multiple time steps, the system becomes increasingly less correlated. These lost of correlations requires entropic dissipations in accordance to the Landauer's Principle.

IV. DISCUSSION

Our study revealed the main result that there is a bound of $k_B T/e$ that cannot be surpassed by any single ratchet design. It has been suggested that this is a consequence of the discontinuous change in joint states during the switching transition [15]. The origins of this results must have come from the costs and dissipations of information processing, as those are the fundamental principles of information processing, and the efficiencies and performance of our ratchets are simply the consequences of them.

These lost of efficiencies are the structural costs of information processing devices that are not captured by Landauer's Principle. They comes from the erasures of correlations or mutual information in the system. The best choices of structural designs for a given information content, is simply the design that minimises the various structural costs of transient and modularity dissipations. An efficient device must minimise these costs, and this is partly done so by careful control of the amount of correlations or mutual information accumulated in the system. This is demonstrated in transient dissipation as dissipations come from the ratchet keeping useless mutual information between itself and the past, and also in modularity dissipation as the erasure of mutual information between subsystems.

There are of course still various other structural costs that were not investigated, understanding of these structural costs can enable better designs of computing devices, that works more efficiently with little dissipations. Insights into the functioning of biological enzymes that utilises Maxwellian work can also be gained, allowing designs of Maxwell's Demon enzyme engines that can fulfils specific tasks that are medical in nature. For example, the information reservoir can have symbols of $\mathcal{Y} \in \{A, T, G, C\}$ for DNA bases, allowing the study of RNA transcoding and correcting errors from mutations, similar to a ratchet working with noisy inputs. Overall, the study of information processing which is how information is created, stored, transferred, shared, erased, and how they are connected to the entropy dissipation in a thermodynamic system, can work towards a physical theory of information.

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Appendix A: Proof of efficiency of memoryless ratchet

A memoryless ratchet cannot create temporal correlations in a process as it cannot store any information from the past in its single ratchet state. Since the input process has no temporal correlations, and the memoryless ratchet cannot create it, the output process must also be memoryless. Formally, we say that the ϵ-transducer of the single state ratchet acts as a memoryless channel for the input process [16], which according to the Data Processing Inequality [13], will reduce the temporal correlations in the input process. Thus, the length-1 redundancy of the output process must be lesser or equal to that of the input process:

$$H' - h'_\mu \leq H[b] - h_\mu \tag{A1}$$

For memoryless inputs, the right hand side is 0, and since length-1 redundancies must be larger or equal to 0, the inequality reduces to an equality:

$$\begin{aligned} H' - h'_\mu &= 0 \\ h'_\mu &= H' \\ &= H[b'] \end{aligned} \tag{A2}$$

Thus, the output process' entropy rate is simply the single-symbol entropy with b' being the statistical bias of the output process. Therefore, according to IPSL, the work production rate of the memoryless ratchet is bounded by:

$$\langle W \rangle_{\text{memoryless}} \leq k_B T \ln 2 \; (H[b'] - H[b]) \tag{A3}$$

On the other hand, for a memoryful ratchet, it is possible to create temporal correlations in the output process as the ratchet can store past information in its ratchet states. This is because the ratchet state that the ratchet is currently in is dependent on past ratchet states, and the output process in turn depends on these ratchet states. Thus, the length-1 redundancy for the output process can be greater than 0:

$$\begin{aligned} H[b'] - h'_\mu &\geq 0 \\ H[b'] &\geq h'_\mu \\ \implies H[b'] - H[b] &\geq h'_\mu - H[b] \end{aligned} \tag{A4}$$

Therefore, according to IPSL, the work production rate of a memoryful ratchet is lesser or equal to the work production rate of a memoryless ratchet:

$$\langle W \rangle_{\text{memoryful}} \leq k_B T \ln 2 \; (h'_\mu - H[b]) \tag{A5}$$

$$\begin{aligned} &\leq k_B T \ln 2 (H[b'] - H[b]) \\ &= \langle W \rangle_{\text{memoryless}} \end{aligned} \tag{A6}$$

Appendix B: Derivation of Transient Dissipation

Continuing from equation 14:

$$\begin{aligned} \frac{\langle Q^{\text{tran}}_{0\to N} \rangle}{k_B T \ln 2} &\geq \lim_{N\to\infty} \left[\frac{\langle Q_{0\to N} \rangle}{k_B T \ln 2} - (-N(h'_\mu - h_\mu)) \right] \\ &\geq H[X_0, Y^G_0] - H[X_N, Y^G_N] + N(h'_\mu - h_\mu) \\ &= H[X_0, Y_{0:\infty}] - H[X_N, Y'_{0:N}, Y_{N:\infty}] \\ &+ N(h'_\mu - h_\mu) \\ &= \big(H[X_0] + H[Y_{0:\infty}] - I[X_0; Y_{0:\infty}]\big) \\ &- \big(H[X_N] + H[Y'_{0:N}, Y_{N:\infty}] - I[X_N; Y'_{0:N}, Y_{N:\infty}]\big) \\ &+ N(h'_\mu - h_\mu) \end{aligned}$$

From the third line to the fourth line, the following identity was used:

$$H[A, B] = H[A] + H[B] - I[A; B]$$

Here, two assumptions will be made. First, we do not want arbitrary increase or decrease in dissipations that comes from different initial conditions—i.e. $H[X_0] - H[X_N] \neq 0$, thus the system is set to start at the stationary state, such that:

$$H[X_0] = H[X_N]$$

Second, the reasonable assumption is that the ratchet is non-anticipatory, such that the ratchet cannot predict or anticipate the incoming tape sequence by itself. This means that the initial ratchet state shares no information with the input tape sequence, or that they have no mutual information between them:

$$I[X_0; Y_{0:\infty}] = 0$$

Upon simplifications and expressing:

$$H[Y'_{0:N}, Y_{N:\infty}] = H[Y'_{0:N}] + H[Y_{N:\infty}] - I[Y'_{0:N}; Y_{N:\infty}],$$

$$H[Y_{0:\infty}] = H[Y_{0:N}] + H[Y_{N:\infty}]$$

most terms will cancels and gives:

$$\begin{aligned}\frac{\langle Q^{\text{tran}}_{0\to N}\rangle}{k_B T \ln 2} &\geq H[Y_{0:N}] - I[Y_{0:N}; Y_{N:\infty}] - H[Y'_{0:N}] \\ &+ I[Y'_{0:N}; Y_{N:\infty}] + I[X_N; Y'_{0:N}, Y_{N:\infty}] \\ &+ N(h'_\mu - h_\mu) \\ &= (H[Y_{0:N}] - Nh_\mu) - I[Y_{0:N}; Y_{N:\infty}] \\ &- (H[Y'_{0:N}] - Nh'_\mu) \\ &+ I[Y'_{0:N}; Y_{N:\infty}] + I[X_N; Y'_{0:N}, Y_{N:\infty}]\end{aligned}$$

The limits of $N \to \infty$ are then taken on both sides, and defining the first and the third terms as the excess entropies of the input and output processes:

$$\begin{aligned}E &= \lim_{N\to\infty} \left(H[Y_{0:N}] - Nh_\mu\right) \\ &= \lim_{N\to\infty} I[Y_{0:N}; Y_{N:\infty}]\end{aligned} \tag{B1}$$

$$\begin{aligned}E' &= \lim_{N\to\infty} \left(H[Y'_{0:N}] - Nh'_\mu\right) \\ &= \lim_{N\to\infty} I[Y'_{0:N}; Y'_{N:\infty}]\end{aligned} \tag{B2}$$

and writing the dissipation as an asymptotic rate:

$$\langle Q^{\text{tran}}\rangle = \lim_{N\to\infty} \langle Q^{\text{tran}}_{0\to N}\rangle$$

we arrived at:

$$\frac{\langle Q^{\text{tran}}\rangle}{k_B T \ln 2} \geq -E' + I[Y'_{0:N}; Y_{N:\infty}] + I[X_0; Y'_{0:N}, Y_{N:\infty}]$$

$$\therefore \frac{\langle Q^{\text{tran}}\rangle_{\min}}{k_B T \ln 2} = -E' + I[Y'_{0:N}; Y_{N:\infty}] + I[X_0; Y'_{0:N}, Y_{N:\infty}] \tag{B3}$$

For the third term, recall that this is a non-anticipatory ratchet:

$$\begin{aligned}I[X_0; Y_{N:\infty}] &= 0 \\ \implies I[X_0; Y'_{0:N}, Y_{N:\infty}] &= I[X_N; Y'_{0:N}]\end{aligned}$$

Note that we also relabelled X_0 as X_N. The second term in equation B3 vanishes as the output's past $Y'_{0:N}$ comes from the input's past $Y_{0:N}$, and has no correlations with the input's future $Y_{N:\infty}$, which are not processed by the ratchet yet.

$$\implies I[Y'_{0:N}; Y_{N:\infty}] = 0 \tag{B4}$$

Therefore, we arrived at:

$$\frac{\langle Q^{\text{tran}}\rangle_{\min}}{k_B T \ln 2} = I[X_N; Y'_{0:N}] - E' \tag{B5}$$

Appendix C: Derivation of Modularity Dissipation

First, we define the joint state of a system at time t to be Z_t. The joint system (global) can be broken down into two subsystems, the interacting (local) subsystem Z^i_t, and the stationary subsystem Z^s_t. The interacting subsystem is the subsystem that is undergoing changes, while the stationary subsystem remains unchanged. This is what it means to make modular changes to a system.

Therefore, the interactions can be expressed in terms of stationary, local, and global interactions:

$$M^{\text{stationary}}_{z^s_t \to z^s_{t+\tau}} = \delta_{z^s_t, z^s_{t+\tau}} \tag{C1}$$

$$M^{\text{local}}_{z^i_t \to z^i_{t+\tau}} = \Pr(Z^i_{t+\tau} = z^i_{t+\tau} \mid Z^i_t = z^i_t) \tag{C2}$$

$$M^{\text{global}}_{(z^i_t, z^s_t) \to (z^i_{t+\tau}, z^s_{t+\tau})} = M^{\text{local}}_{z^i_t \to z^i_{t+\tau}} \delta_{z^s_t, z^s_{t+\tau}} \tag{C3}$$

Whereby the identity operator is used for the stationary interaction, such that $z^s_t = z^s_{t+\tau}$.

To derive the dissipation due to such a modular interaction, we start from the average entropy production of the universe $\langle\Sigma\rangle$ in an interval τ, which is defined as the work done in that interval, minus away the change in non-equilibrium free energy F^{neq}_t:

$$\langle\Sigma_{t\to t+\tau}\rangle = \frac{1}{T}\left[\langle W_{t\to t+\tau}\rangle - (F^{\text{neq}}_{t+\tau} - F^{\text{neq}}_t)\right] \tag{C4}$$

The non-equilibrium free energy is defined as the weighted sum of the joint states' internal energy U_z minus by the entropies of these joint states.

$$F^{\text{neq}}_t = \sum_z \Pr(Z_t = z) U_z - k_B T \ln 2 \; H[Z_t] \tag{C5}$$

Taking all states to have the same internal free energy and substituting the expression for F^{neq}_t into equation C4, we have the generalised Landauer's Principle [14]:

$$\langle\Sigma_{t\to t+\tau}\rangle = \frac{\langle W_{t\to t+\tau}\rangle}{T} + k_B \ln 2 \left(H[Z_{t+\tau}] - H[Z_t]\right) \tag{C6}$$

We should emphasise at this point that the work done $\langle W_{t\to t+\tau}\rangle$ is not the work production of the system that we have been discussing in the past chapters. Rather, this is the work required to be put into the system in order to do computation. Thus, in the context of our information ratchet, this is actually the heat $\langle Q\rangle$ absorbed from the heat reservoir.

Now, modularity will be applied. The minimum work done of a modular computation is simply the minimum amount of work required to do computation or information processing at the local subsystem, and is negative of the change in entropy of the interacting or local subsystem. Note that this is simply the generalised information processing bound as in equation 12.

$$\begin{aligned}\langle W_{t\to t+\tau}\rangle &\geq \langle W^{\text{local}}_{t\to t+\tau}\rangle_{\min} \\ &= k_B T \ln 2 \left(H[Z^i_t] - H[Z^i_{t+\tau}]\right)\end{aligned} \tag{C7}$$

On the other hand, if we were to do computation globally without modularity, the minimum required work is negative the change in entropy of the entire system.

$$\langle W^{\text{global}}_{t\to t+\tau}\rangle_{\min} = -k_B T \ln 2 \left(H[Z_{t+\tau}] - H[Z_t]\right) \tag{C8}$$

Upon substituting equations C7 and C8 into equation C6, we find that the minimum dissipation or entropy increase due to modular information processing or computation is:

$$T\langle \Sigma^{\text{mod}}_{t\to t+\tau}\rangle_{\min} = \langle W^{\text{local}}_{t\to t+\tau}\rangle_{\min} - \langle W^{\text{global}}_{t\to t+\tau}\rangle_{\min} \tag{C9}$$

Thus, we can see that the modularity dissipation is simply the additional amount of work required to do a computation modularly. If we were to do the computation globally, we have $\langle W_{t\to t+\tau}\rangle = \langle W^{\text{global}}_{t\to t+\tau}\rangle_{\min}$, and there will be no modularity dissipation. Simplifying equation C9, we arrived at the general equation for modularity dissipation:

$$\begin{aligned}\frac{\langle \Sigma^{\text{mod}}_{t\to t+\tau}\rangle_{\min}}{k_B \ln 2} &= \frac{\langle W^{\text{local}}_{t\to t+\tau}\rangle_{\min}}{k_B T \ln 2} + H[Z^i_{t+\tau}, Z^s_t] - H[Z^i_t, Z^s_t] \\ &= H[Z^i_t] - H[Z^i_{t+\tau}] + H[Z^i_{t+\tau}, Z^s_t] \\ &\quad - H[Z^i_t, Z^s_t] + (H[Z^s_t] - H[Z^s_t]) \\ &= I[Z^i_t; Z^s_t] - I[Z^i_{t+\tau}; Z^s_t]\end{aligned} \tag{C10}$$

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